

A Polynomial-Time 1/4-Competitive Algorithm for Edge-Arrival Secretary Matching

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Abstract

We study the edge-arrival secretary matching problem in a general weighted graph. The edges arrive in a uniformly random order, their weights are arbitrary and are revealed on arrival, and the online algorithm must irrevocably decide whether to accept each edge while maintaining a matching. Assuming the number m of edges is known, we give a simple polynomial-time deterministic online algorithm whose expected weight is at least

$$\frac{\lfloor m/2 \rfloor \lceil m/2 \rceil}{m(m-1)} \text{OPT} \geq \frac{1}{4} \text{OPT} \quad (m \geq 2),$$

and which is optimal for $m = 1$ in the trivial sense. The algorithm rejects the first half of the stream and thereafter accepts an arriving edge whenever it belongs to a fixed tie-broken maximum-weight matching of the edges observed so far and its endpoints are still free. The main point of the proof is a reverse-exposure lemma showing that, conditional on an edge $e = uv$ arriving at time t , the probability that both u and v remain available is at least

$$\prod_{i=\lfloor m/2 \rfloor + 1}^{t-1} \left(1 - \frac{2}{i}\right).$$

Thus this note gives a solution, not merely partial progress, to the polynomial-time implementation question for the 1/4 edge-arrival guarantee in the model of Ezra, Feldman, Gravin, and Tang, subject to the standard assumption that m is known. It does not improve the constant beyond 1/4 and does not address variants in which m is unknown.

1 Introduction

Secretary matching in general graphs combines two constraints: uncertainty about future arrivals and the irrevocability of matching decisions. In the edge-arrival version, an adversarially weighted edge set is presented in a uniformly random order; when an edge arrives, its weight is revealed, and the algorithm must either accept it immediately, provided this preserves feasibility of the current matching, or reject it forever. Ezra, Feldman, Gravin, and Tang [1] gave a 1/4-competitive edge-arrival result, but the algorithmic description involves conditional endpoint-availability probabilities. The computational issue is whether one can obtain the same 1/4 guarantee by a polynomial-time online rule without evaluating those probabilities.

The purpose of this note is to record a direct affirmative answer. The algorithm is almost as simple as the classical secretary cutoff rule. It samples the first half of the edges. After the cutoff, when edge e_t arrives, it computes the maximum-weight matching $M^*(E_t)$ in the graph induced by the observed edge set E_t . If $e_t \in M^*(E_t)$ and both endpoints of e_t are still unmatched, it accepts e_t ; otherwise it rejects e_t .

The analysis does not attempt to compute the exact conditional probability that the endpoints of e_t are available. Instead, it lower bounds this probability by exposing the previously observed edges in reverse order. At a reverse step with i remaining observed edges, at most two of those i edges can both lie in the current maximum matching and touch the two endpoints currently under consideration. This gives a multiplicative survival bound. Summing the resulting round-by-round contribution gives exactly the desired $1/4$ guarantee.

Status of the result. Under the standard edge-arrival secretary matching model with known m , Theorem 1 is a full solution to the polynomial-time $1/4$ -competitive-algorithm question. It is not merely partial progress. The result should not be read as a better-than- $1/4$ competitive ratio, nor as a treatment of unknown-horizon variants.

2 Model and notation

Let $G = (V, E)$ be an undirected graph with $m = |E|$ edges. The value of m is known to the algorithm. Edge weights $(w_e)_{e \in E}$ are fixed adversarially. They may be taken to be real numbers, with the convention that the empty matching is feasible; hence no maximum-weight matching ever needs to contain a negative-weight edge. Equivalently, one may restrict attention to nonnegative weights.

The edges arrive in a uniformly random order

$$e_1, e_2, \dots, e_m.$$

At time t , the algorithm observes the identity of e_t , its endpoints, and its weight w_{e_t} , and then must irrevocably accept or reject it. Accepted edges must always form a matching. Let

$$E_t = \{e_1, \dots, e_t\}$$

be the set of edges observed by time t .

For every edge set $S \subseteq E$, let $M^*(S)$ denote one fixed maximum-weight matching in the graph (V, S) . Ties are resolved by a deterministic rule depending only on S , the weights of edges in S , and fixed edge labels, and not on the arrival order by which S was revealed. For instance, among maximum-weight matchings one may choose the lexicographically first incidence vector with respect to a fixed ordering of all edge labels. Let

$$\mu^* = M^*(E), \quad \text{OPT} = w(\mu^*) = \sum_{e \in \mu^*} w_e.$$

If $\text{OPT} = 0$, all inequalities below are trivial, so the interesting case is $\text{OPT} > 0$.

3 The algorithm

Set

$$r = \left\lfloor \frac{m}{2} \right\rfloor.$$

The algorithm is deterministic, apart from the random order of the input stream.

Algorithm 1: Current-optimum cutoff algorithm

1: Initialize the accepted matching $\mu \leftarrow \emptyset$ and the set $A \leftarrow V$ of currently available vertices.

- 2: For $t = 1, 2, \dots, m$, observe edge $e_t = uv$ and weight w_{e_t} .
- 3: If $t \leq r$, reject e_t .
- 4: If $t > r$, compute $M_t = M^*(E_t)$.
- 5: If $t > r$, $e_t \in M_t$, and $u, v \in A$, accept e_t by setting

$$\mu \leftarrow \mu \cup \{e_t\}, \quad A \leftarrow A \setminus \{u, v\}.$$

Otherwise reject e_t .

- 6: After round m , return μ .

The algorithm is online because it uses only the observed edge set E_t , observed weights, the fixed tie-breaking rule, and the known value of m . It runs in polynomial time because a maximum-weight matching in a general graph can be computed in polynomial time [2], and the algorithm performs at most one such computation per round.

4 A reverse-exposure survival lemma

For $t > r$, define

$$\beta_t = \prod_{i=r+1}^{t-1} \left(1 - \frac{2}{i}\right),$$

where the empty product is 1. When $r \geq 2$, this product simplifies to

$$\beta_t = \prod_{i=r+1}^{t-1} \frac{i-2}{i} = \frac{r(r-1)}{(t-1)(t-2)}. \quad (1)$$

Lemma 1 (Two-endpoint survival). *Fix $t > r$, a set $S \subseteq E$ with $|S| = t$, and an edge $e = uv \in S$. Condition on the event*

$$\mathcal{C}_{S,e} = \{E_t = S \text{ and } e_t = e\}.$$

Under Algorithm 1, the probability that both u and v are available at the beginning of round t is at least β_t :

$$\mathbb{P}[u, v \text{ available before round } t \mid \mathcal{C}_{S,e}] \geq \beta_t.$$

Proof. Throughout the proof, all probabilities are conditional on $\mathcal{C}_{S,e}$. Let

$$Q = S \setminus \{e\}, \quad |Q| = t - 1.$$

Given $\mathcal{C}_{S,e}$, the edges of Q occupy the first $t - 1$ positions in a uniformly random order.

Expose this order backwards. Start with $Q_{t-1} = Q$. For $i = t - 1, t - 2, \dots, 1$, choose f_i uniformly from Q_i and set

$$Q_{i-1} = Q_i \setminus \{f_i\}.$$

In the corresponding forward order, f_i is precisely the edge arriving at time i , and the set of edges observed by time i is $Q_i = \{f_1, \dots, f_i\}$.

For each exploitation round $i \in \{r + 1, \dots, t - 1\}$, define the dangerous set

$$D_i(Q_i) = \{g \in M^*(Q_i) : g \text{ is incident to } u \text{ or } v\}.$$

Since $M^*(Q_i)$ is a matching, it contains at most one edge incident to u and at most one edge incident to v . Hence

$$|D_i(Q_i)| \leq 2. \quad (2)$$

Define the good event

$$A_i = \{f_i \notin D_i(Q_i)\}.$$

We now justify the product lower bound carefully. Let $\mathcal{F}_{i+1}$ be the sigma-algebra generated by the later reverse exposures $f_{i+1}, \dots, f_{t-1}$, with $\mathcal{F}_t$ trivial. Conditional on $\mathcal{F}_{i+1}$, the set

$$Q_i = Q \setminus \{f_{i+1}, \dots, f_{t-1}\}$$

is determined and has cardinality i , while f_i is uniform over Q_i . Therefore, by (2),

$$\mathbb{P}(A_i \mid \mathcal{F}_{i+1}) = 1 - \frac{|D_i(Q_i)|}{i} \geq 1 - \frac{2}{i}. \quad (3)$$

Moreover, for $j > i$, the event A_j is $\mathcal{F}_{i+1}$ -measurable. Applying (3) successively from $i = t - 1$ down to $i = r + 1$ gives

$$\mathbb{P}\left(\bigcap_{i=r+1}^{t-1} A_i\right) \geq \prod_{i=r+1}^{t-1} \left(1 - \frac{2}{i}\right) = \beta_t.$$

It remains only to connect the events A_i to availability. During rounds $1, \dots, r$, Algorithm 1 rejects every edge. During an exploitation round $i \in \{r + 1, \dots, t - 1\}$, the algorithm can accept an edge incident to u or v only if the arriving edge f_i belongs to $M^*(Q_i)$ and is incident to u or v , i.e., only if $f_i \in D_i(Q_i)$. Thus, on the event $\bigcap_{i=r+1}^{t-1} A_i$, neither u nor v is matched before round t . Hence both endpoints are available at the beginning of round t . The preceding probability bound proves the lemma. $\square$

5 Competitive analysis

Theorem 1. *For every weighted graph with $m \geq 2$ edges, Algorithm 1 satisfies*

$$\mathbb{E}[w(\mu)] \geq \frac{\lfloor m/2 \rfloor \lceil m/2 \rceil}{m(m-1)} \text{OPT} \geq \frac{1}{4} \text{OPT}.$$

For $m = 1$, the same rule with $r = 0$ obtains the offline optimum.

Proof. The case $m = 1$ is immediate, so assume $m \geq 2$. Let W_t be the weight accepted by Algorithm 1 in round t , with $W_t = 0$ if no edge is accepted. For $t \leq r$, $W_t = 0$. Fix a round $t > r$.

Condition on $E_t = S$ and $e_t = e$. If $e \notin M^*(S)$, the algorithm does not accept e at time t . If $e \in M^*(S)$, the algorithm accepts e whenever both endpoints of e are available. By Lemma 1, for every S and $e \in S$,

$$\mathbb{E}[W_t \mid E_t = S, e_t = e] \geq \beta_t w_e \mathbf{1}[e \in M^*(S)]. \quad (4)$$

Here the right-hand side is nonnegative: a maximum-weight matching, with the empty matching feasible, contains no negative-weight edge.

Now condition only on $E_t = S$. Given the unordered set $E_t = S$, the last edge e_t is uniform over the t edges of S . Averaging (4) over this uniform choice gives

$$\mathbb{E}[W_t \mid E_t = S] \geq \frac{\beta_t}{t} \sum_{e \in S} w_e \mathbf{1}[e \in M^*(S)] \quad (5)$$

$$= \frac{\beta_t}{t} w(M^*(S)). \quad (6)$$

Taking expectation over the random t -set E_t yields

$$\mathbb{E}[W_t] \geq \frac{\beta_t}{t} \mathbb{E}[w(M^*(E_t))]. \quad (7)$$

Since $\mu^* \cap E_t$ is a feasible matching in (V, E_t) ,

$$w(M^*(E_t)) \geq w(\mu^* \cap E_t).$$

The set E_t is a uniformly random t -element subset of E , so each edge of μ^* belongs to E_t with probability t/m . Therefore

$$\mathbb{E}[w(M^*(E_t))] \geq \mathbb{E}[w(\mu^* \cap E_t)] = \frac{t}{m} \text{OPT}. \quad (8)$$

Combining (7) and (8), we obtain

$$\mathbb{E}[W_t] \geq \frac{\beta_t}{m} \text{OPT}. \quad (9)$$

Summing over all exploitation rounds gives

$$\mathbb{E}[w(\mu)] = \sum_{t=r+1}^m \mathbb{E}[W_t] \geq \frac{\text{OPT}}{m} \sum_{t=r+1}^m \beta_t. \quad (10)$$

It remains to evaluate the sum. If $r \geq 2$, then by (1),

$$\begin{aligned} \sum_{t=r+1}^m \beta_t &= r(r-1) \sum_{t=r+1}^m \frac{1}{(t-1)(t-2)} \\ &= r(r-1) \sum_{t=r+1}^m \left(\frac{1}{t-2} - \frac{1}{t-1} \right) \\ &= r(r-1) \left(\frac{1}{r-1} - \frac{1}{m-1} \right) \\ &= \frac{r(m-r)}{m-1}. \end{aligned}$$

For $m = 2$ and $m = 3$, where $r = 1$, the same identity follows directly from the product definition: the sums are respectively 1 and 1. Thus, for every $m \geq 2$,

$$\mathbb{E}[w(\mu)] \geq \frac{r(m-r)}{m(m-1)} \text{OPT} = \frac{\lfloor m/2 \rfloor \lceil m/2 \rceil}{m(m-1)} \text{OPT}.$$

If $m = 2k$ is even, the coefficient equals

$$\frac{k^2}{(2k)(2k-1)} = \frac{k}{2(2k-1)} \geq \frac{1}{4}.$$

If $m = 2k + 1$ is odd, the coefficient equals

$$\frac{k(k+1)}{(2k+1)(2k)} = \frac{k+1}{2(2k+1)} \geq \frac{1}{4}.$$

This proves the theorem. $\square$

Corollary 1. *In the known- m edge-arrival secretary matching model, there is a polynomial-time online algorithm with competitive ratio at least $1/4$ for general weighted graphs.*

6 Why this bypasses the conditional-probability obstacle

The algorithm analyzed here is not an implementation of the conditional-probability algorithm of [1]. In particular, it never computes an exact probability that the endpoints of the current edge are still available. It simply accepts every feasible arriving edge that belongs to the fixed current optimum matching.

The proof works because exact availability probabilities are unnecessary. Fix the edge $e = uv$ under analysis and expose the earlier observed edges backwards. When there are i possible choices for the next reverse-exposed edge, the set of choices that could immediately match u or v under Algorithm 1 has size at most two: they are the edges of the current maximum matching incident to u or v . This statement is deterministic after the current reverse-exposed set Q_i is fixed. Hence the conditional hazard is at most $2/i$ at every reverse step, even though the exact future evolution of the algorithm may be complicated. The survival probability is therefore bounded below by the product of the complements of these hazards. This lower bound is precisely strong enough to sum to a $1/4$ competitive ratio.

7 Limitations and assumptions

The theorem should be interpreted with the following assumptions.

- The number m of edges is known, as in the usual finite-horizon secretary model and in the edge-arrival formulation considered here.
- The tie-breaking rule for $M^*(S)$ is fixed independently of the random arrival order. This is essential for the reverse-exposure proof, because $D_i(Q_i)$ must be determined by Q_i before the uniformly random choice of f_i is made.
- The result matches the $1/4$ edge-arrival guarantee but does not improve the competitive ratio.
- The proof gives an online polynomial-time algorithm for the edge-arrival model. It does not address different arrival models, adversarial-order online matching, or unknown-horizon variants.

References

- [1] Tomer Ezra, Michal Feldman, Nikolai Gravin, and Zhihao Gavin Tang. Tight bounds for secretary matching in general graphs. *Mathematics of Operations Research*, 50(4):3039–3054, 2025. Earlier version: *Secretary Matching with General Arrivals*, arXiv:2011.01559.
- [2] Jack Edmonds. Paths, trees, and flowers. *Canadian Journal of Mathematics*, 17:449–467, 1965.